CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

May 2014

CL 207 Concrete Technology

Date: 02 .05 .2014, Friday **Time: 10:00 a.m. To 1:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Design a concrete mix (as per IS: 10262-2009) for a reinforced concrete work which will be exposed to the Server condition. The concrete mix is to be designed as data given below. Refer useful given in Table 1,2,3 & 5 of corresponding IS. **11**

Stipulations for Mix design :

- Grade of designation: M 35
- Types of cement : OPC 53 grade
- Maximum nominal size of aggregate : 10 mm
- Workability in terms of Slump: 125 mm
- Method of concrete placing: Pumping
- Degree of supervision : Good
- Type of aggregate: Crushed angular aggregate
- Maximum cement content: 450 kg/ m^3
- Chemical admixture type: Superplasticiser (Use 0.7% of total cementitious material content)

Test data of materials (As per IS):

- Specific gravity of: Coarse aggregate: 2.70, Fine aggregate: 2.67, Cement: 3.15 and Chemical admixture: 1.12 .
- Water absorption (IS 2386:1963) : (i) Coarse aggregate: 0.5% and (ii) Fine aggregate: 1 %.
- Free (surface) moisture: (i) Coarse aggregate: Nil (absorbed moisture also nil) (ii) Fine aggregate: nil.
- Sieve analysis (IS 2386 Part 1): (i) Coarse aggregate: (Conforming to Table 2 of IS 383) and (ii) Fine aggregate: Conforming to grading Zone III.

Q - 2 (a) What is heat of hydration? Draw rate of heat libration during process of hydration of cement with time. Why it become necessary to study and controls it? **04**

(b) What is soundness of cement and how is it tested? **03**

(c) What are the advantages of using Ready mixed concrete instead of in-situ mixed concrete? Consider material, execution and practical aspect. **05**

OR

Q - 2	(a)	Describe the chemical compounds of Portland Cement.	04
	(b)	Explain briefly the dry manufacturing process of Portland Cement.	03
	(c)	What are the properties of materials used for mass concrete?	05
Q - 3	(a)	Explain advantages of Portland Pozzolona Cement. Where you will prefer its use?	03
	(b)	Write short note on Sulphate resisting cement including chemical composition, its use and advantages.	04
	(c)	State your observations regarding future trends and new developments in concrete technology.	05
		OR	
Q - 3	(a)	Differentiate between rapid hardening and quick setting cement.	04
	(b)	Explain High Alumina Cement with its characteristics.	03
	(c)	What are the constituents of concrete? State their relative proportions in ordinary concrete.	05

SECTION - II

Q - 4	(a)	Explain with example, how to interpret the result of Slump test of workability of fresh concrete.	03
	(b)	List the parameters responsible for corrosion in R.C.C..	03
	(c)	If Sodium sulphate present in water is 100 ppm, will you accept water for concreting? Justify your answer.	01
Q - 5	(a)	Explain deflocculating effect of plasticizer on cement grain which is responsible for water reduction in concrete.	03
	(b)	State two name of air entraining agent. Also state effect of air entrainment on the properties of concrete.	04
	(c)	Explain phenomenon of bulking of fine aggregate. What will be its effect in weight batching?	03
	(d)	Write experimental procedure of determination of aggregate crushing value of coarse aggregate. State permissible value for its acceptance in construction work.	04
		OR	
Q – 5	(a)	State effect of superplasticizer on the properties of Harden concrete.	04
	(b)	What is GGBS? State source of it and its advantages using in concrete.	03
	(c)	Explain how shape, size and texture of coarse aggregate affect w/c ratio of concrete.	04
	(d)	In the years to come natural sand will not be available in large quantity for big Infrastructural projects. What will alternative solution of this problem as per you?	03
Q – 6	(a)	State factors affecting workability of fresh concrete with relevant explanation.	07
	(b)	State parameters induce micro cracks in harden concrete. How micro cracks reducing durability of concrete?	07
		OR	
Q – 6	(a)	Compaction and Curing are most important steps in concrete. Why? Explain defects cause in concrete due improper Compaction and Curing.	07

(b) State factors affecting compressive strength of concrete. Explain effect of w/c ration in detail. 07

Table 1 Assume Standard Derivation (IS 10262 :2009, Clauses 3.2.1.2, A-3 and B-3,Page-2)		
Sr.No.	Grade of Concrete	Assume Standard Derivation N/mm^2
1	M 10	3.5
2	M 15	
3	M 20	4.0
4	M 25	
5	M 30	5.0
6	M 35	
7	M 40	
8	M 45	
9	M 50	
10	M 55	
Note: The above values correspond to site control having proper storage of cement; weigh batching of all materials; controlled addition of water; regular checking of all materials, aggregate grading and moisture content; and periodical checking of workability and strength. Where there is deviation from the above, values given in the above table shall be increased by $1 N/mm^2$		

Table 2 Maximum Water Content per Cubic Metre of Concrete for Nominal Maximum Size of Aggregate (IS 10262 :2009, Clauses 4.2,A-5 and B-5, Page-3)		
Sr.No.	Nominal Maximum size of aggregate (mm)	Maximum Water Content # kg
1	10	208
2	20	186
3	40	165
Note: These quantities of mixing water are use in computing cementitious material contents for trial batches. #Water content corresponding to saturated surface dry aggregate		

Table 3. Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate (IS 10262 :2009, Clauses 4.4,A-7 and B-7, Page-3)					
Sr.No.	Nominal Maximum size of aggregate (mm)	Volume of coarse aggregate# per unit volume of total aggregate for different zones of fine Aggregate (For water-cement ratio = 0.5)			
		Zone IV	Zone III	Zone II	Zone I
1	10	0.50	0.48	0.46	0.44
2	20	0.66	0.64	0.62	0.6
3	40	0.75	0.73	0.71	0.69
#Volumes are based on aggregates in saturated surface dry condition.					

Table : 5 Minimum Cement Content,Maximum Water -Cement Ratio and Minimum Grade of Concrete for Diffenebt Exposure with Weight Aggregartes of 20 mm Nominal Maximum Size (IS 456- 200,Clauses 6.1.2.,8.2.4.1.and 9.1.2.,Page -20)							
Sr. No	Exposure	Plain Concrete			Reinforcement Concrete		
		Minimum Cement Content kg /m³	Maximum Free Water - Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg /m³	Maximum Free Water - Cement Ratio	Minimum Grade of Concrete
i	Mild	220	0.60	--	300	0.55	M 20
ii	Moderate	240	0.60	M 15	300	0.50	M 25
iii	Servere	250	0.50	M 20	320	0.45	M 30
iv	Very Servere	260	0.45	M 20	340	0.45	M 35
v	Extreme	280	0.40	M 25	360	0.40	M 40

Note :

1.Cement content prescribed in the above table is irrespective of the grades of cement and it is inclusive of additions mentioned in 5.2.The additions such as flyash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and w/c ratio if suitability established and as iong as the maximum amounts taken into account do not exceed the limit of pozzolana and slag specified in IS 1489(partt) and IS 455 respectively

2. Minimun grade for plain concret under mild exposure condition is not spcified.

******